$$(-2)^{\overline{n}} = \sum_{k=0}^{n} (-2)^k \begin{bmatrix} n \\ k \end{bmatrix}$$

$$\|$$

ale $n \geq 3 \Rightarrow \quad -2^{\overline{n}} = 0$

$$\frac{1}{(1-x)^m} = \sum_{n=0}^{\infty} \binom{n+m-1}{m-1} x^n$$

$$x^{\underline{n}} = \sum_{k=0}^{n} (-1)^{n-k} \begin{bmatrix} n \\ k \end{bmatrix} x^k$$

$$x^n = \sum_{k=0}^{n} \begin{Bmatrix} n \\ k \end{Bmatrix} x^{\underline{k}}$$

$$(-x)^n = \sum_{k=0}^{n} \begin{Bmatrix} n \\ k \end{Bmatrix} (-x)^{\underline{k}} = \sum_{k=0}^{n} \begin{Bmatrix} n \\ k \end{Bmatrix} (-1)^k x^{\overline{k}}$$

$$\Rightarrow \quad x^n = \sum_{k=0}^{n} \begin{Bmatrix} n \\ k \end{Bmatrix} (-1)^{n-k} x^{\overline{k}}$$

$$x^{\overline{n}} = \sum_{k=0}^{n} \begin{bmatrix} n \\ k \end{bmatrix} x^k$$

$$(-x)^{\overline{n}} = \sum_{k=0}^{n} \begin{bmatrix} n \\ k \end{bmatrix} (-x)^k$$

$$(-1)^n x^{\underline{n}} = \sum_{k=0}^{n} \begin{bmatrix} n \\ k \end{bmatrix} (-1)^k x^k$$

$$x^{\underline{n}} = \sum_{k=0}^{n} \begin{bmatrix} n \\ k \end{bmatrix} (-1)^{n-k} x^k$$

$\sigma \in S_n$

$$I_\sigma = \{ j \in \operatorname{dom} \sigma \mid \forall_{i \in [j-1]} \sigma(i) < \sigma(j) \}$$

$$A_k = \{ \sigma \mid |I_\sigma| = k \}$$

$$T: \quad |A_k| = \begin{bmatrix} n \\ k \end{bmatrix}$$

$$A_{n,k} = \{ \sigma \in S_n \mid |I_\sigma| = k \}$$

jeśli $k > n \begin{bmatrix} n \\ k \end{bmatrix} = |A_{n,k}| = 0$

zatem niech $n = 0 \quad k = 0 \quad |A_{n,k}| = \begin{bmatrix} n \\ k \end{bmatrix} = 1$

i niech $n = 1 \quad k = 0 \quad |A_{n,k}| = 0 = \begin{bmatrix} n \\ k \end{bmatrix}$

i niech $n = 1 \quad k = 1 \quad |A_{n,k}| = 1 = \begin{bmatrix} n \\ k \end{bmatrix}$

$$|A_{n,k}| \overset{?}{=} (n-1) |A_{n-1,k}| + |A_{n-1,k-1}|$$

$$\begin{bmatrix} n \\ k \end{bmatrix} = (n-1) \begin{bmatrix} n-1 \\ k \end{bmatrix} + \begin{bmatrix} n-1 \\ k-1 \end{bmatrix}$$

$$\sigma^{-1}(n) =: m$$

$$m = n \Rightarrow \sigma|_{[n-1]} \in A_{n-1,k-1}$$

$$m < n \Rightarrow \varphi \in S_{n-1} \quad \varphi|_{[n-1]\setminus\{m\}} = \sigma|_{[n-1]\setminus\{m\}}$$

$$\left(C_{2n} \right) \qquad C(x) = \quad x\, C(x)^2 + 1$$

$$\frac{1-\sqrt{1-4x}}{2x}$$

$$\frac{\dfrac{1-\sqrt{1-4x}}{2x} + \dfrac{1-\sqrt{1+4x}}{-2x}}{2}$$

$$= \quad \frac{\sqrt{1+4x} - \sqrt{1-4x}}{4x}$$

$$A(x) + A(-x) = \sum a_n x^n + a_n (-x)^n$$

$$= \sum 2 a_{2n} x^{2n}$$

$$B(x) = \frac{A(x) + A(-x)}{2} = \sum \quad a_{2n} x^{2n}$$

$$B(\sqrt{x}) = \frac{A(\sqrt{x}) + A(-\sqrt{x})}{2} = \sum a_{2n} x^n$$

$$\frac{\sqrt{1+4\sqrt{x}} - \sqrt{1-4\sqrt{x}}}{4\sqrt{x}}$$

$$\frac{\sqrt{x+4x\sqrt{x}} - \sqrt{1-4x\sqrt{x}}}{4x}$$

$$a_{n+2} = 6a_{n+1} - 9a_n + 3^n$$

$$A(x) = \sum_{n=0}^{\infty} a_n x^n = a_0 + a_1 x + \sum_{n=0}^{\infty} a_{n+2} x^{n+2}$$

$$= 6\sum_{n=0}^{\infty} a_{n+1} x^{n+2} - 9\sum_{n=0}^{\infty} a_n x^{n+2} + \sum_{n=0}^{\infty} 3^n \cdot x^{n+2}$$

$$= 6x\,A(x) - 9x^2 A(x) + \frac{x^2}{1-3x}$$

$$(1-6x+9x^2)A(x) = \frac{x^2}{1-3x}$$

$$A(x) = \frac{x^2}{(1-3x)(1-3x)^2} = \frac{x^2}{(1-3x)^3}$$

$$a_n = A\cdot 3^n + Bn3^n + Cn^2 3^n$$

$a_0 = 0 \Rightarrow \quad A = 0$

$a_1 = 0 \Rightarrow \quad B + C = 0$

$a_2 = \ 3$

$18B - 36B = 1$

$B = \frac{-1}{18}$

$\binom{k}{k} + \binom{k+1}{k}$

$$\binom{0}{0}$$

$$\binom{1}{0} \quad \binom{1}{1}$$

$$\binom{2}{0} \quad \binom{2}{1} \quad \binom{2}{2}$$

$$\binom{3}{0} \quad \binom{3}{1} \quad \binom{3}{2} \quad \binom{3}{3}$$

$$\binom{4}{0} \quad \binom{4}{1} \quad \binom{4}{2} \quad \binom{4}{3} \quad \binom{4}{4}$$

$$\binom{5}{0} \quad \binom{5}{1} \quad \binom{5}{2} \quad \binom{5}{3} \quad \binom{5}{4} \quad \binom{5}{5}$$

$$x^{\overline{n}} = \sum^{n} \left[{n \atop k}\right] \cdot x^{k}$$

$k=0$

$$x^n = \sum_{k=0}^{n} \left\{ {n \atop k} \right\} x^{\underline{k}}$$